

Quanta Scout OS — Unconstrained Vision

Additions, Substitutions, and Full-Scale Architecture

*This document covers what Quanta Scout OS would look like with no budget constraints — full production channels, upgraded AI stack, native mobile apps, and enterprise-grade infrastructure. Each section marks whether it is an **Addition** (new capability) or **Substitution** (replacing something currently in the system).*

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1. Scout Input Channels

1.1 Telegram Bot — Full Two-Way Integration

Type: Substitution (replaces simulated OpenClaw) **Cost driver:** WhatsApp Business API (\$0.005–\$0.08/conversation) + OpenClaw or custom gateway

The current system has OpenClaw wired at the gateway layer but messages are simulated. In production:

- Scouts submit leads by **texting a Quanta Telegram bot** — no app, no portal, no friction
- Message arrives → Intake Agent structures it → bot replies with one follow-up question → scout answers → deal is created
- Bot handles follow-up reminders, check-ins, and questions from the Quanta team — all in the same Telegram thread the scout already uses
- Conversation state is maintained per-scout so a scout can send fragmented updates over multiple days and they are stitched together correctly

Implementation:

```
Scout texts: "Met a founder building AI for trucking dispatch, 3 pilots, raising $500K seed"
Bot replies: "Interesting. What's their background – have they worked in logistics before?"
Scout texts: "Yeah, ex-Delhivery ops. Name is Arjun Mehta."
Bot replies: "Got it. Submitted to Quanta. I'll follow up if they need more info."
```

1.2 WhatsApp Business API

Type: Addition **Cost driver:** Meta WhatsApp Business API (~\$0.005–\$0.08/conversation) + verified business number

WhatsApp is the dominant messaging channel for scouts in India. The exact same flow as Telegram — but scouts never leave their primary app.

- Requires Meta Business Verification and a dedicated WhatsApp number
 - Template messages for check-ins (Meta requires pre-approved templates for outbound)
 - Free-form replies work within 24-hour conversation windows
 - Attachments (pitch decks, voice notes) handled natively
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1.3 Slack App

Type: Addition **Cost driver:** Slack app listing + Slack API (free for basic, Slack Connect for enterprise)

For scouts embedded in VC ecosystems, accelerators, or tech communities who already live in Slack:

- `/quanta submit` slash command → opens a modal for quick submission
 - Bot DMs the scout follow-up questions
 - Quanta team gets a `#new-deals` channel with real-time deal cards
 - Partner can react with 🔥 (high priority) or ❌ (pass) directly in Slack to update deal status
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1.4 SMS Fallback

Type: Addition **Cost driver:** Twilio SMS (~\$0.0079/message)

For scouts in low-connectivity or low-smartphone regions:

- Basic text-based intake: scout texts a number, structured bot conversation collects the deal
 - No rich formatting — plain text Q&A
 - Works on any phone, any network
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1.5 Email-Native Submission

Type: Addition **Cost driver:** Mailgun or SendGrid inbound parsing (free tier available)

Scout forwards a founder's email or their own notes to `submit@scout.quantaventures.com` :

- Inbound email parsed → AI extracts deal details from the body
 - Attachments (PDFs, decks) automatically processed by the enrichment agent
 - Reply thread becomes the deal conversation
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2. AI Stack Upgrades

2.1 Web Research & Live Enrichment

Type: Substitution (replaces document-only enrichment) **Cost driver:** Exa API (\$5/1,000 searches) or Perplexity API (\$5/1,000 requests)

Current enrichment only runs on uploaded documents. With a live search API:

- After every submission, automatically search the web for the startup, founders, and market
- Pull: LinkedIn profiles, Crunchbase data, AngelList presence, news mentions, domain traffic estimates (SimilarWeb), GitHub activity for technical founders
- Cross-reference what the scout said with what is publicly available — flag discrepancies
- Auto-populate founder backgrounds without the scout needing to provide LinkedIn URLs

Prompt pattern:

```
Search: "[Startup Name] [Founder Name] startup"
Extract: funding rounds, team size, press coverage, product URLs, customer logos
Flag: if public data contradicts scout's claims (e.g., scout says "pre-seed" but
Crunchbase shows Series A)
```

2.2 Semantic Duplicate Detection with pgvector

Type: Substitution (replaces text-similarity duplicate check) **Cost driver:** OpenAI Embeddings API (\$0.0001/1K tokens) + pgvector Supabase extension (free)

Current duplicate detection uses string matching on startup names and descriptions. This misses:

- Same company submitted under different names ("FlowOps" vs "Flow Operations")
- Semantically identical companies ("AI for trucking dispatch" and "logistics automation using ML")

With embeddings:

- Every deal description is embedded at submission time and stored in `deal_embeddings` table
- Cosine similarity search against all existing embeddings
- Threshold: >0.92 similarity → flag as likely duplicate
- Show matched deal to Quanta team with side-by-side comparison

Schema addition:

```
alter table deals add column embedding vector(1536);
create index deals_embedding_idx on deals using ivfflat (embedding
vector_cosine_ops);
```

2.3 GPT-4o Vision — Pitch Deck from Photo

Type: Addition **Cost driver:** GPT-4o API (~\$0.01/1K input tokens for vision)

Scout photographs a slide deck, whiteboard, or product screenshot with their phone camera:

- Image sent to GPT-4o vision endpoint
 - Extracts: company name, problem statement, solution, team slide, traction metrics, market size claims
 - Structures into the same deal schema as manual entry or voice pitch
 - Works for handwritten notes, napkin sketches, or physical whiteboards
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2.4 Real Whisper API — Dedicated Endpoint

Type: Substitution (replaces Groq shared Whisper) **Cost driver:** OpenAI Whisper API (\$0.006/minute) or AssemblyAI (\$0.0025/minute)

Current transcription uses Groq's free-tier Whisper which has:

- 28,800 audio seconds/day shared limit
- 25MB file size limit
- No speaker diarization

Production upgrade:

- Dedicated Whisper endpoint (OpenAI or AssemblyAI) with no shared limits
 - **Speaker diarization:** separate scout voice from any other voices in a recording
 - **Timestamps:** link specific transcript segments to specific questions
 - **AssemblyAI add-ons:** auto-chapters, key phrases, sentiment per segment, filler word detection
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2.5 Founder LinkedIn Auto-Fetch

Type: Addition **Cost driver:** PhantomBuster or Proxycurl API (\$49–\$300/month)

Scout enters a LinkedIn URL → system automatically pulls:

- Full name, headline, current company
- Previous companies and tenures
- Education (university, degree, year)
- Number of connections (proxy for network strength)
- Recent posts / activity (proxy for thought leadership)

Auto-populates the `founders` table and pre-fills the Q&A answer for founder background.

2.6 Voice Conviction Scoring

Type: Addition **Cost driver:** Groq / Assembly AI (already in stack)

Beyond transcription — analyse HOW the scout talks about a deal:

- **Specificity score:** does the scout cite actual numbers, names, and dates? Or is it vague?
- **Confidence markers:** count of hedging language ("I think", "maybe", "not sure") vs. conviction language ("absolutely", "best I've seen", "must look at this")
- **Coherence:** does the pitch flow logically from problem → solution → why now → traction?

Output added as a `voice_conviction_score` signal card alongside existing VC signals.

2.7 Structured Deal Comparison ("Comps Intelligence")

Type: Addition **Cost driver:** Exa/Perplexity + OpenAI GPT-4o

When a deal is submitted, automatically:

1. Search for 3–5 comparable companies that raised in similar space + stage
 2. Pull their funding amounts, investors, and outcomes
 3. Present to Quanta team as a "How does this compare?" widget in the deal overview
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3. Scout Experience

3.1 React Native Mobile App

Type: Substitution (replaces mobile web) **Cost driver:** React Native development + App Store / Play Store (\$25 one-time + \$99/year)

Current scout experience is a mobile-optimised web app. A native app would add:

- **Background audio recording:** record a pitch while driving without keeping the screen on
- **Offline-first:** submit a deal with no internet — syncs automatically when connected
- **Push notifications:** "Quanta asked you a question about NeuralCart" — scout replies directly from the notification

- **Camera integration:** photograph a pitch deck or business card → auto-processed
- **Widgets:** home screen widget showing pending follow-ups
- **Haptic feedback:** satisfying "submitted" confirmation

Tech stack: React Native + Expo + Supabase Realtime

3.2 WhatsApp-Native Onboarding

Type: Addition

Current onboarding: Quanta team sends invite email → scout clicks link → creates account on web.

WhatsApp-native flow:

1. Quanta team sends a WhatsApp message to the scout's number
2. Scout replies "yes" → bot sends a one-time PIN
3. Scout enters PIN → account created
4. All submissions happen in WhatsApp from that point

Zero web friction. No email needed.

3.3 Smart Re-engagement Messages

Type: Addition

When a scout replies "nothing this week" to a check-in, instead of ending there:

- AI checks the scout's focus areas and past submissions
 - Sends a **targeted prompt**: "What about the HealthTech space? You mentioned hospital-facing tools last month — anything new?"
 - Or references a specific founder they mentioned earlier: "Any update from the Meesho alum you were meeting?"
 - Uses GPT-4o with the scout's full history as context
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3.4 Scout Leaderboard & Gamification

Type: Addition

Scout portal home page shows:

- Personal "Scout Score" based on quality and volume of submissions
- Rank within the network (anonymised)
- Streak: consecutive weeks with at least one signal
- Badges: "First Submission", "High Signal Scout", "5 Deals Under Review"

Not gamification for its own sake — gives scouts a tangible sense of their contribution to Quanta's deal flow.

4. Quanta Team Experience

4.1 Quanta CoPilot Sidebar

Type: Addition **Cost driver:** GPT-4o + Exa web search + pgvector

While reviewing a deal, a persistent sidebar AI that answers questions using:

- Your full deal history

- Portfolio company data
- Live web search

Example queries:

- "How does this compare to our logistics bets?"
- "What's the market size for agri-fintech in Maharashtra?"
- "Have we seen any similar founders before?"
- "What did we learn from our last SaaS investment that applies here?"

4.2 Partner Weekly Digest Email

Type: Addition **Cost driver:** Resend + GPT-4o (already in stack)

Every Monday morning, an AI-generated digest to Evan and Mateo:

This week in your deal flow:

NEW DEALS (3)

- NeuralCart – B2B SaaS, Seed · Strong founder, 25% MoM growth · Scout: Atharva
- AgriPulse – AgriTech, Pre-seed · Early signal · Missing: pitch deck
- HealthSync – HealthTech, Series A · Under review

FOLLOW-UPS DUE (2)

- Rahul Patil (FlowOps) promised pitch deck by May 25
- Priya Menon (NeuralCart) – awaiting pilot customer names

STATUS CHANGES

- CarbonLoop → Intro Requested (moved by Mateo)
- DevPulse → Monitor (no update in 3 weeks)

SUGGESTED ACTIONS

- NeuralCart: all signals green – consider scheduling a call
- AgriPulse: ask scout for deck before spending more time

4.3 One-Click Intro Email Drafting

Type: Addition **Cost driver:** GPT-4o (already in stack)

On any deal detail page, a "Draft Intro" button:

- GPT writes a warm intro email from Evan/Mateo to the founder
- References specific deal details, scout context, and Quanta's investment thesis
- Scout conviction included naturally ("Our scout Atharva has been tracking this space for months...")
- Partner reviews and sends in one click

4.4 Supabase Realtime Deal Feed

Type: Substitution (replaces polling) **Cost driver:** Supabase Realtime (included in Supabase Pro plan)

Current inbox refreshes on page load. With Realtime:

- New deal arrives → immediately appears in inbox with a subtle animation
- Scout sends a message → conversation thread updates live without refresh

- Status changes visible to all team members simultaneously
 - "Evan is viewing this deal" presence indicators
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4.5 Portfolio Conflict Detection

Type: Addition

When a new deal is submitted, automatically check against:

- Existing portfolio companies (competitive overlap)
- Existing deal pipeline (duplicate outreach risk)

Output: warning banner on the deal card — "Similar to [Portfolio Co] — check competitive dynamics before proceeding."

5. Data & Intelligence Layer

5.1 pgvector Semantic Deal Search

Type: Addition **Cost driver:** pgvector extension (free on Supabase) + OpenAI Embeddings

Replace the current keyword search with semantic search:

- "Show me deals similar to AgriPulse" → finds all deals with similar problem, market, and stage
 - "What B2B SaaS deals are in our pipeline?" → semantic matching, not just category tags
 - Quanta team can search by concept: "founder with deep ops background in logistics" → finds relevant deals
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5.2 Scout Performance Analytics

Type: Addition

Per-scout dashboard visible only to Quanta team:

- Signal quality ratio: % of submissions that reach "Under Review" or better
- Category hit rate: which sectors is this scout strongest in?
- Response rate: % of check-ins responded to within 48 hours
- Submission cadence: weekly, monthly, seasonal trends
- Comparative rank: anonymised benchmarking within the network

Used for: coaching scouts, deciding which scouts to activate for specific deal categories, rewarding top performers.

5.3 Market Theme Clustering

Type: Addition **Cost driver:** GPT-4o + pgvector

Automatically clusters deals by emerging market theme:

- Detects when multiple scouts independently surface deals in the same space
 - Generates a "Market Pulse" note: "3 scouts flagged climate logistics deals in 6 weeks — this may be a structural shift worth tracking"
 - Surfaces to Quanta team as a thematic alert
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5.4 Signal Drift Alerts

Type: Addition **Cost driver:** Exa web search + cron job (already in stack)

For every deal in the pipeline, weekly automated check:

- Did this startup raise a round publicly? (Crunchbase, TechCrunch)
- Did the founder get press coverage?
- Did they launch a product (Product Hunt, LinkedIn post)?
- Did a competitor raise?

If yes → auto-flag the deal and update the record. Partner receives a notification.

5.5 CRM Export & Integration

Type: Addition **Cost driver:** Salesforce / HubSpot API (varies)

One-click export of any deal or scout record to:

- Salesforce (for VCs already using it as a portfolio tracker)
 - HubSpot (deal pipeline)
 - Notion (as a structured database)
 - Airtable (for team collaboration)
 - CSV export for any report
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6. Infrastructure & Reliability

6.1 Background Job Queue (Inngest or Trigger.dev)

Type: Substitution (replaces fire-and-forget pipeline calls) **Cost driver:** Inngest (\$0 free tier → \$50/month Pro)

Current `runPostSubmissionPipeline` is a fire-and-forget Promise — if it fails, it fails silently with no retry.

With Inngest:

- Every pipeline step is a tracked, retryable job
 - Full observability: see exactly which step failed, with the error and input
 - Automatic exponential backoff retries (3× by default)
 - Fan-out: Step 2A (signals), 2B (brief), and 2C (enrichment) run as truly parallel jobs, not concurrent promises
 - Failure alerting: Slack / email notification when a job fails after all retries
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6.2 Upstash Redis — Rate Limiting & Idempotency

Type: Addition **Cost driver:** Upstash Redis (\$0 free tier → \$10/month)

- **Rate limiting:** prevent scouts from submitting 50 deals in 5 minutes (accidental or abuse)
 - **Webhook idempotency:** OpenClaw may deliver the same message twice — Redis deduplication prevents duplicate deals
 - **Session caching:** cache `user_roles` lookup per user for 5 minutes — eliminates DB query on every middleware call
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6.3 Dedicated Vercel Edge Config

Type: Addition **Cost driver:** Vercel Pro (\$20/month)

- **Edge Config:** store feature flags, rate limits, and killswitches at the edge — changes take effect in <100ms globally
- **Edge Middleware analytics:** see exactly where scouts and team members are dropping off
- **Preview deployments:** every PR gets a full preview URL with isolated DB (via Supabase branching)

6.4 Supabase Branching for Safe Migrations

Type: Addition **Cost driver:** Supabase Pro (\$25/month)

- Each feature branch gets its own Supabase branch (isolated DB)
- Test schema changes and seed data without touching production
- Merge branch → migration applied automatically on merge to main
- Eliminates the risk of "I ran a migration in prod by accident"

6.5 Monitoring & Alerting (Sentry + Datadog)

Type: Addition **Cost driver:** Sentry Developer (\$26/month) + Datadog APM (\$15/host/month)

- **Sentry:** captures every client-side and server-side exception with full stack trace, breadcrumbs, and session replay
- **Datadog APM:** traces every API request end-to-end — see exactly which DB query is slow, which AI call timed out
- **Custom alerts:** if `/api/startup/init` errors exceed 5% in 10 minutes → PagerDuty alert

7. Full Stack Comparison Table

Layer	Current (Demo Build)	Unconstrained Vision
Scout channels	Web app only	Web + Telegram + WhatsApp + Slack + SMS + Email-in
Transcription	Groq Whisper (shared, 25MB limit)	Dedicated Whisper / AssemblyAI (unlimited, diarization)
LLM	Groq Llama (free tier)	GPT-4o + Groq hybrid (GPT-4o for precision, Groq for speed)
Enrichment	Uploaded docs only	Live web search (Exa/Perplexity) + LinkedIn auto-fetch + vision
Duplicate detection	Text similarity	pgvector semantic embeddings (cosine similarity)
Mobile	Mobile web	React Native app (iOS + Android)
Realtime	Polling	Supabase Realtime WebSockets
Email	Resend shared domain	Resend verified domain (any recipient) + weekly digest

Pipeline jobs	Fire-and-forget Promise	Inngest background jobs (retries, observability)
Rate limiting	None	Upstash Redis edge rate limiting
Monitoring	Console logs	Sentry + Datadog APM
DB branching	Manual migrations	Supabase Branching per PR
Search	None / keyword	pgvector semantic search
Analytics	Basic counts	Scout performance scoring, market clustering, signal drift
Integrations	None	Salesforce / HubSpot / Notion / Airtable export
Auth	Cookie-based	Supabase Auth + SAML SSO for enterprise Quanta team

Priority Order for Next Build Phases

If budget unlocks incrementally, this is the recommended build order by ROI:

Priority	Feature	Why First
1	Telegram / WhatsApp bot	Highest scout engagement lift — removes all friction from submission
2	React Native app	Voice pitch quality improves dramatically on native audio APIs
3	Inngest job queue	Reliability — pipeline failures are invisible right now
4	Live web enrichment (Exa)	Every deal gets richer automatically — no extra scout effort
5	pgvector semantic search + duplicate detection	Prevents duplicate outreach and surfaces deal patterns
6	Partner weekly digest	Highest Quanta team leverage — turns the system into a CIO-level briefing
7	Supabase Realtime	Removes the "why isn't this updating" confusion
8	Scout analytics + leaderboard	Retention and quality improvement for the scout network
9	GPT-4o vision (deck photos)	High convenience for scouts who receive physical decks
10	CRM integrations	Enables Quanta to bridge Scout OS with existing portfolio management